Asymmetric Mamba–CNN Collaborative Architecture for Large-Size Remote Sensing Image Semantic Segmentation

Jinbo Zhang, Min Chen, Yitao Zhao, Lianlei Shan, Caiyi Li, Han Hu, *Member, IEEE*, Xuming Ge, Qing Zhu, and Bo Xu

Abstract*— Large-size remote sensing images contain rich geographical information. Efficient and accurate semantic segmentation of these images is of significant importance in various fields. However, the massive memory requirements have hindered the development of semantic segmentation methods for large-size remote sensing images. Most existing methods struggle to balance memory usage, global modeling, and local representation accuracy. To address these issues, we propose a new semantic segmentation method for large-size remote sensing images, Mamba–CNN parallel network (MCPNet), which demonstrates impressive performance. The method is an asymmetric Mamba–convolutional neural network (CNN) hybrid architecture. Given the linear modeling complexity of Mamba, we construct the M-branch based on the visual state space (VSS) model, which processes downsampled images to reduce memory consumption while alleviating Mamba's local forgetting problem. To further enhance the model's capability in fine-grained detail extraction, we meticulously design a detail-preserving network (DPN) as the C-branch. This branch employs a split downsampling strategy and multiscale convolutional kernel groups to process large-size images, ensuring the preservation of spatial positional relationships while capturing fine-grained local details. Moreover, to effectively filter redundant information introduced by large-size images and bridge the semantic gap between the features extracted by CNN and Mamba, we propose a multigated feature fusion module (MG-FFM). This module progressively refines heterogeneous feature alignment through a bottom-up hierarchical refinement strategy, achieving a progressive fusion of semantics and details. Our method achieves state-of-the-art (SOTA) performance in terms of mean intersection over union (mIoU) and mF1 score on the self-constructed Yaan UAV dataset and two widely used public datasets (DeepGlobe and Inria Aerial) while consuming less GPU memory. The codes will be available at *https://github.com/fsqy-zhang/MCPNet

***Index Terms*— Convolutional neural network (CNN), feature fusion, large-size image, Mamba, semantic segmentation.**

NOMENCLATURE

SOTA	State of the art.
DPN	Detail-preserving network.
SSM	State space model.
CNN	Convolutional neural network.
VSS	Visual state space.
MG-FFM	Multigated feature fusion module.
CE loss	Cross-entropy loss.
mse loss	Mean squared error loss.
GSD	Ground sampling distance.
OOM	Out of memory.
Tr.	Training.
Inf.	Inference.
Mem.	Memory.
Src.	Source domain.
Tgt.	Target domain.

I. INTRODUCTION

WITH the advancement of remote sensing technology, the availability of large-size remote sensing images has significantly improved. Semantic segmentation of remote sensing images has attracted great attention in both computer vision and remote sensing communities, playing an essential role in various fields such as urban planning [1], agricultural management [2], environmental monitoring [3], and disaster response [4].

Compared with semantic segmentation tasks on natural and medical images, remote sensing images exhibit richer spatial interactions between objects. Effectively leveraging the abundant contextual information in remote sensing images is critical for semantic segmentation tasks. However, due to GPU memory limitations, conventional semantic segmentation methods struggle to directly process large-size remote sensing images. Common approaches include either downsampling images or dividing them into smaller patches before feeding them into the model. The former inevitably results in significant detail loss, while the latter limits the local receptive field of each patch, fundamentally constraining

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the accessible contextual information. Moreover, the slicing process can cause object truncation, leading to incomplete information that may negatively impact model performance [Fig. 1(a)].

To tackle this challenge, some studies [13], [14], [16], [17], [18] have adopted coarse-to-fine optimization strategies or global–local collaborative frameworks to reduce memory consumption while improving semantic segmentation accuracy. However, multistage processing and excessive image partitioning significantly degrade inference speed. Yu et al. [12], [19] utilize lightweight dual-branch feature extraction networks to directly process large-size images, yet their simple convolutional layers fail to effectively capture fine details. Guo et al. [15] and Ji et al. [20] introduce Laplacian pyramids to enhance high-frequency features while using dilated convolutions and atrous spatial pyramid pooling (ASPP) to expand the receptive field. However, limited by the modeling approach, they still exhibit deficiencies in global modeling.

Transformer-based models leverage self-attention mechanisms to effectively capture global dependencies and establish long-range relationships, achieving more robust semantic responses. These models have been widely adopted in image processing tasks [6], [21], [22]. However, as the input sequence length increases, the computational complexity of Transformer-based models grows quadratically, making them inefficient in handling ultralarge remote sensing images due to high computational and memory demands.

Recently, the structured SSM (S4) [23], by introducing the high-order polynomial projection operator (HiPPO) [24], has significantly enhanced the model's ability to capture long-range dependencies. Based on S4, Mamba integrates selective scanning mechanisms and hardware-aware algorithms, achieving success in natural language processing tasks. Its linear complexity and exceptional global modeling capability have attracted widespread attention in image processing and remote sensing research [7], [25], [26], [27]. Among them, RS-Mamba [7] is the first to apply SSM to the remote sensing domain for directly processing large-size images. It employs a UNet-like structure and is trained on an A100 GPU, demonstrating the potential of the Mamba architecture in handling large-size images. However, the model lacks structural optimization for memory efficiency. Under resource-limited conditions (e.g., a single RTX 4090 or GPUs with smaller memory), both training and inference are severely hindered. As shown in Fig. 1(b), RS-Mamba requires 11 260 MB of memory to directly infer a single 2432 × 2432 image, which is approximately four times that of our method [Mamba–CNN parallel network (MCPNet)].

Overall, the current methods still have the following issues.

1) Conventional semantic segmentation models are often constrained by GPU memory, forcing them to either downsample images or process image patches. The former inevitably leads to significant loss of detail, while the latter provides only a limited local receptive field within each patch, resulting in substantial loss of contextual information. Additionally, the slicing process often causes object truncation, which may convey incomplete information and negatively impact model performance, as illustrated in Fig. 1(a).
2) Methods designed for large-size image semantic segmentation and certain lightweight models can directly handle large-size images while preserving rich contextual information. However, they either require multistage data processing, leading to a complex training process and low data processing efficiency, or are constrained by their architectures, making it difficult for them to learn sufficiently rich and complex feature representations. Consequently, these methods struggle to balance global modeling and local representation accuracy, ultimately affecting model performance, as illustrated in Fig. 1(c).

These limitations highlight the need for innovative solutions capable of efficiently processing large-size remote sensing images while preserving their rich contextual information. To address this challenge, we propose MCPNet, an asymmetric Mamba–CNN collaborative architecture for large-size remote sensing image semantic segmentation. Our goal is to achieve a better tradeoff between memory consumption and accuracy in this task. Unlike hybrid architectures such as [11], [28], and [29], which process the original image using parallel, same-scale encoders with multilevel feature interactions, leading to excessive memory consumption, MCPNet introduces a differentiated architecture. It consists of a shallow CNN-based branch and a deep Mamba-based branch, each specializing in modeling images at different scales. The CNN-based shallow branch processes large-scale images to retain local details, while the Mamba-based deep branch operates on downsampled images, efficiently capturing global dependencies while mitigating Mamba's tendency to forget local details. Additionally, the decoding stage employs a bottom-up strategy, where the rich semantics from the Mamba branch guide the CNN branch in detail restoration. This scale-differentiated hybrid design enables MCPNet to process large-size images with lower memory consumption while achieving superior performance. Moreover, we introduce a dynamic contextual training strategy, allowing the model to adaptively acquire multiscale contextual information during training while reducing memory requirements. Thanks to its scalability, we adopt a large-patch slicing inference strategy, which mitigates the inference speed bottleneck caused by excessive slicing in previous methods [13], [14].

Furthermore, we develop a detail-preserving network (DPN) as our CNN-based branch, incorporating split sampling and heterogeneous convolution kernel groups to further enhance the model performance. To effectively filter redundant information from large-size images and address feature misalignment caused by the different feature extraction mechanisms of CNNs and Mamba, such as disparities in channel count, feature resolution, feature values, and hierarchical representations, we design an MG-FFM. This module employs a spatial–channel residual multigating mechanism for feature selection and facilitates progressive multilevel feature fusion. Our approach fully leverages the structural advantages of both Mamba and CNN, achieving an optimal balance between accuracy and memory efficiency.

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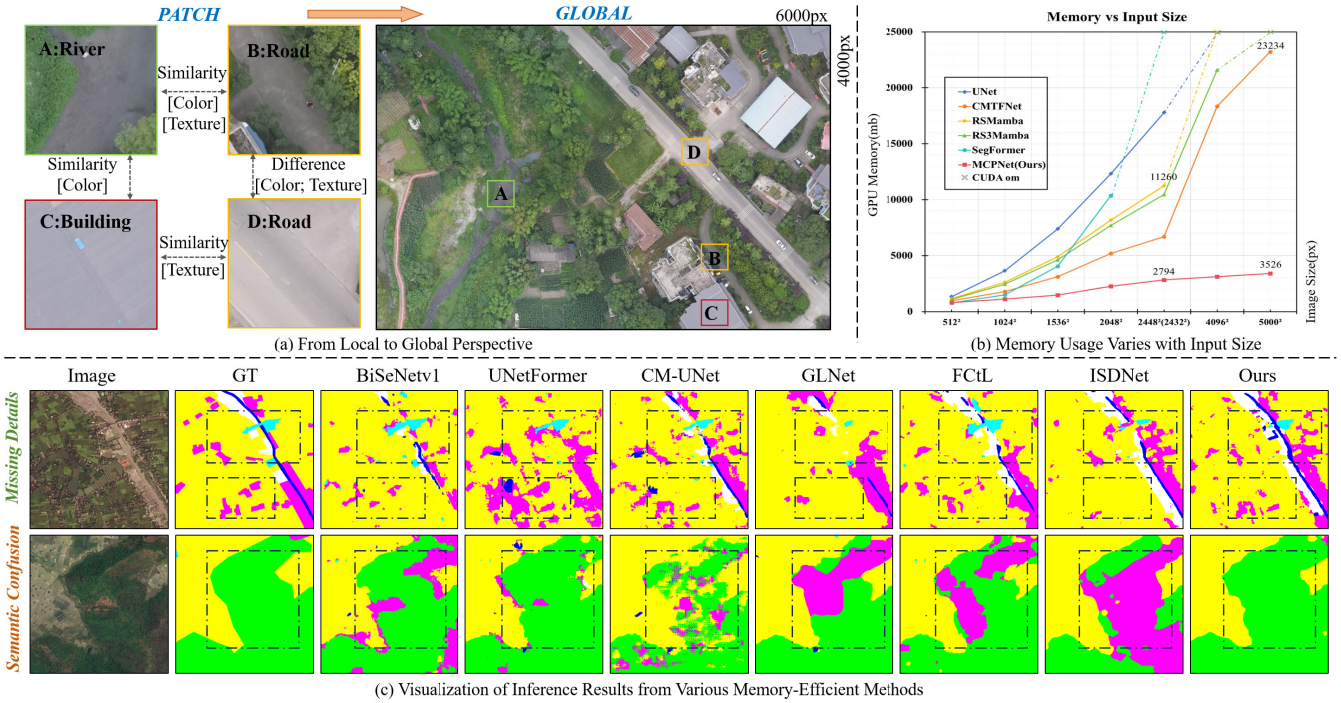


Fig. 1. Challenges in large-size remote sensing image semantic segmentation. (a) Large-size remote sensing images contain richer ground object information, and the slicing operation only captures local receptive fields, fundamentally limiting the available contextual information, which leads to semantic confusion between ground objects. (b) Conventional semantic segmentation methods experience a rapid increase in memory consumption as the input image size grows, making it infeasible to process large-size images directly. Here, we visualize the memory consumption during inference for several representative models with increasing input sizes, including UNet [5] (CNN-based), SegFormer [6] (Transformer-based), RS-Mamba [7] (Mamba-based), RS3Mamba [8], CMTFNet [9] (hybrid architecture), and MCPNet (our method). (c) Some lightweight approaches and methods designed for ultralarge image semantic segmentation are constrained by their network architectures (UNetFormer [10], CM-UNet [11], BiSeNetv1 [12], GLNet [13], FCtL [14], and ISDNet [15]), making it challenging to learn sufficiently rich and complex feature representations. Consequently, they struggle to balance global modeling capability and local representation accuracy.

Specifically, our contributions are as follows.

1) We propose MCPNet, an asymmetric Mamba–CNN collaborative architecture for large-size remote sensing image semantic segmentation. This framework integrates a CNN-based shallow branch to preserve local details and a Mamba-based deep branch for global modeling. During decoding, deep semantics from the Mamba branch guide the CNN branch in progressively restoring high-resolution features, ultimately generating high-resolution semantic segmentation results. Additionally, we introduce a dynamic continuous contextual training strategy, which expands the model's accessible context while reducing memory usage during training.
2) Since the Mamba branch involves downsampling, leading to a significant loss of details, this imposes higher demands on the CNN branch for enhanced detail extraction capabilities. To address this challenge, we carefully design a DPN as the CNN branch of MCPNet. This network utilizes a split sampling strategy and multiscale convolution kernel groups to enhance the model's ability to capture fine details.
3) To effectively filter redundant information from large-scale images and bridge the semantic gap between CNN-branch and Mamba-branch extracted features, we design an MG-FFM. This module employs spatial gating to focus on salient regions, channel gating to suppress modal noise interference, and residual gating to mitigate feature drift during fusion. Through a bottom-up progressive refinement approach, it achieves heterogeneous features alignment and enables semantic-detail progressive fusion.
4) We validate our method on the self-constructed Yaan UAV dataset and two widely used public datasets (DeepGlobe [30] and Inria Aerial [31]), where MCPNet demonstrates superior performance.

II. Related Work

In this section, we first review the research on large-size image semantic segmentation methods. Since our method is a Mamba–CNN hybrid architecture, we next review the application of SSMs in computer vision and research on hybrid architectures.

A. Large-Size Images Semantic Segmentation

Currently, methods for large-size image semantic segmentation can be broadly categorized into two types: 1) progressive refinement of semantic segmentation results and 2) end-to-end processing of large-size images with more detailed network architectures. The first category includes works such as CascadePSP [16] and MagNet [17]. CascadePSP introduces a category-agnostic local boundary refinement module to improve the coarse semantic segmentation results of existing models, generating more precise semantic segmentation

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outputs. MagNet designs a multiscale framework that progressively refines semantic segmentation results through multiple stages.

The second category is represented by milestone works such as global–local networks (GLNet) [13], which pioneered a global–local collaborative architecture for ultrahigh-resolution image semantic segmentation. GLNet employs two branches to separately process global and local features, integrating them through deep feature fusion to obtain semantic segmentation results. FCtL [14] builds a multiscale feature extraction framework and leverages a locality-aware contextual correlation module to adaptively fuse multiscale features. Integrating shallow and deep networks (ISDNet) [15] simultaneously feeds high-frequency residual features and downsampled images into shallow and deep branches, avoiding the need for local feature slicing and significantly accelerating inference speed. Similarly, wide-context network (WoSiNet) [32] constructs two CNN-based encoders and selectively projects contextual information onto local features via a context Transformer. As an improvement over ISDNet, wavelet-based shallow and deep networks (WSDNet) [20] integrates multilevel discrete wavelet transform (DWT) and inverse DWT (IWT), preserving more spatial details in the deep branch while eliminating relation-aware feature fusion (RAF), thereby enhancing semantic segmentation accuracy while improving inference speed. Additionally, some real-time semantic segmentation methods [12], [19], [33], due to their lightweight architectural designs, can also directly process large-scale images. However, these methods are inherently limited by their model architectures, making it difficult for them to learn sufficiently rich and complex feature representations. As a result, they struggle to balance global modeling and local representation accuracy, ultimately affecting overall performance.

B. SSM for Visual Applications

Recently, SSMs have emerged as a promising architecture for sequence modeling tasks. Compared to Transformers, SSMs and their variants offer a global perspective with linear complexity [23], [24]. Building on this, Mamba [34] introduces an input-dependent parameterization method, further optimized through hardware-aware algorithms, significantly enhancing SSM performance while enabling efficient training and inference. This advancement has driven its widespread adoption in sequence-based tasks.

As Mamba has been successfully extended to general vision tasks through Vim [35] and VMamba [36], it is now widely applied across various computer vision domains, including medical image semantic segmentation [37], [38], super-resolution reconstruction [39], [40], low-light enhancement [41], [42], video processing [43], [44], and 3-D vision [37], [45].

Meanwhile, Mamba has also attracted increasing attention in the remote sensing community. MambaHSI [46] constructs spatial and spectral Mamba blocks for hyperspectral image (HSI) classification, revealing its potential as a next-generation backbone for HSI models. CDMamba [47] applies Mamba to change detection in remote sensing imagery and achieves excellent performance. FreMamba [48] introduces a frequency-assisted Mamba framework for remote sensing image super-resolution. Slice Mamba [49] employs horizontally and vertically sliced Mamba blocks for multisource remote sensing data fusion and classification. RS-Mamba [7] was the first to apply SSMs to large-size remote sensing images, demonstrating Mamba's potential for such tasks. However, their model lacks memory-efficient design, making training and inference on large-size images challenging on memory-constrained devices.

C. Hybrid Architecture

A well-designed and efficient hybrid architecture can typically provide more comprehensive feature representations and has long been a research focus in the field of computer vision. Studies [50], [51] have established bidirectional bridges between multiple CNN and Transformer blocks to facilitate the mutual fusion of local and global features. Li et al. [52], Srinivas et al. [53], and Dai et al. [54] have integrated convolution with self-attention mechanisms to enhance performance and improve generalization capability. Meanwhile, some studies [55], [56], [57] have combined Transformer and Mamba frameworks to improve computational efficiency while further strengthening the model's capacity for global interaction understanding.

Recently, several studies have explored the integration of CNNs, Transformers, and Mamba architectures in remote sensing semantic segmentation. Wu et al. [9], Wang et al. [10], and Wu et al. [58] combine convolution-based encoders with Transformer-based decoders to enhance long-range modeling capabilities. CTCFNet [29] employs a parallel, scale-consistent encoder to extract and integrate complementary global and local information from remote sensing images. RS3Mamba [8] and CM-UNet [11], based on the U-Net architecture, fuse CNNs with Mamba for semantic segmentation of remote sensing images, incorporating multiscale connections to capture fine-grained semantic segmentation details.

However, these hybrid architectures primarily focus on feature interactions across different architectures, often overlooking the issues of excessive information redundancy and significant memory overhead introduced by large-size inputs. As a result, these methods struggle to effectively process large-size remote sensing images.

III. Method

This section introduces the detailed composition of MCPNet through five subsections. First, Section III-A provides an overview of the overall framework. Then, Sections III-B–III-D present the dynamic context training strategy, the parallel heterogeneous feature extraction process, and the structural details of MG-FFM, respectively. Finally, the optimization process employed during model training is discussed (Section III-E).

A. Overview

In this article, we propose a network called MCPNet for the semantic segmentation task of large-size remote sensing images, as shown in Fig. 2.

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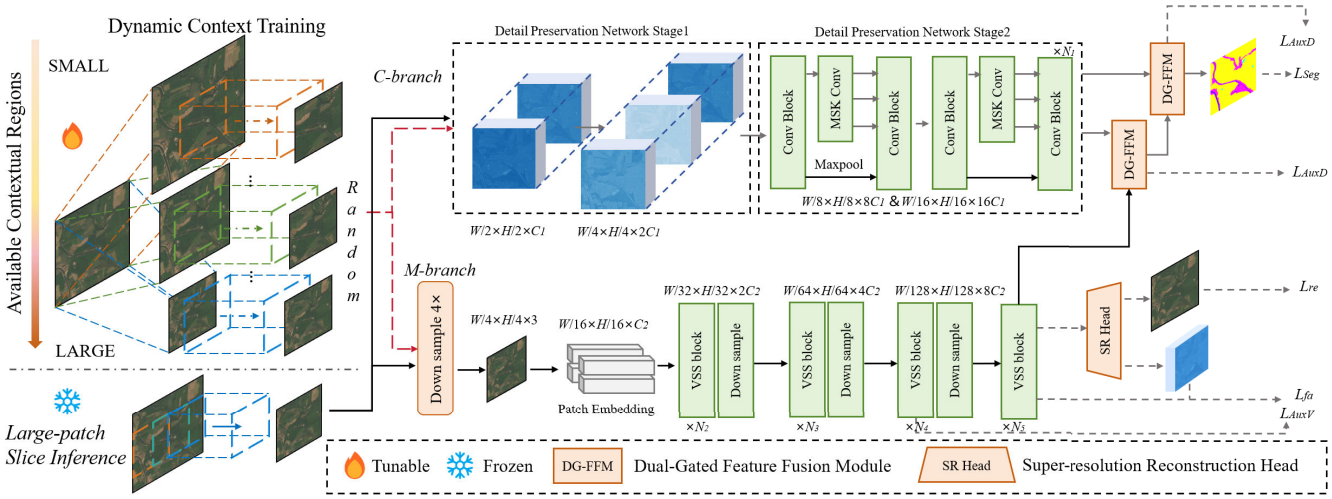


Fig. 2. Pipeline of the proposed MCPNet. Given an image $\mathbf{I} \in \mathbb{R}^{W\times H\times 3}$, the training phase employs a dynamic context sampling strategy to generate $\mathbf{I}_{dy} \in \mathbb{R}^{k\times k\times 3}$, which is then processed by the detail preserving network in the C-branch to extract local detail features. Meanwhile, the downsampled $\mathbf{I}_m \in \mathbb{R}^{(W/4)\times(H/4)\times 3}$ is fed into the M-branch, which consists of an SSM to capture global dependencies. Subsequently, a cascaded MG-FFM module progressively aggregates the features from both branches and restores the high-resolution semantic segmentation results. During training, the M-branch maintains the spatial structure through an SR Head. In this figure, the arrows indicate the data flow path, while the dashed arrows represent processes that are included only during training.

Algorithm 1 Training and Inference Process of MCPNet

Input: Input image $\mathbf{I} \in \mathbb{R}^{W\times H\times 3}$; context parameters α, β, k; optional cropping window $[n, m]$ and stride s for inference
Output: Semantic segmentation output $\mathcal{O}_{Seg}$; auxiliary outputs for training

if Inference mode and $W \times H > 6.25 \times 10^6$ **then**
  $\mathbf{I}_{in} \leftarrow \text{CropWindow}(\mathbf{I}, [n, m], s)$
else $\mathbf{I}_{in} \leftarrow \mathbf{I}$
if Training mode **then**
  Dynamic context generation: $\mathbf{I}_{dy} \leftarrow \lceil\rfloor(\mathbf{I}_{in}; \alpha, \beta, k)$
else $\mathbf{I}_{dy} \leftarrow \mathbf{I}_{in}$
Multi-level global features:
  $\mathbf{I}_m \leftarrow \text{Downsample}(\mathbf{I}_{dy})$
  $\{\mathbf{I}_{m1}, \mathbf{I}_{m2}, \mathbf{I}_{m3}, \mathbf{I}_{m4}\} \leftarrow \mathcal{F}_g(\mathbf{I}_m)$ ▷ Eq. (6)–(8)
Multi-level local features:
  $\{\mathbf{I}_{dp1}, \mathbf{I}_{dp2}, \mathbf{I}_{dp3}, \mathbf{I}_{dp4}\} \leftarrow \mathcal{F}_l(\mathbf{I}_{dy})$ ▷ Eq. (9)–(12)
Heterogeneous feature fusion:
  $\mathbf{F}_{fusion1} \leftarrow \phi(\mathbf{I}_{m4}, \mathbf{I}_{dp4})$
  $\mathbf{F}_{fusion2} \leftarrow \phi(\mathbf{F}_{fusion1}, \mathbf{I}_{dp3})$ ▷ Eq. (13)–(19)
Final decoding:
  $\mathcal{O}_{Seg} \leftarrow \mathcal{D}_{Seg}(\mathbf{F}_{fusion2})$
if Training mode **then**
  $\mathcal{O}_{AuxVss} \leftarrow \mathcal{D}_{AuxVss}(\mathbf{I}_{m3})$
  $\mathcal{O}_{AuxDec} \leftarrow \mathcal{D}_{AuxDec}(\mathbf{F}_{fusion2})$
  $\mathcal{O}_{Re}, \mathcal{O}_{aff} \leftarrow \mathcal{D}_{Re}(\mathbf{I}_{m4})$
  return $\mathcal{O}_{Seg}, \mathcal{O}_{AuxVss}, \mathcal{O}_{AuxDec}, \mathcal{O}_{Re}, \mathcal{O}_{aff}$
else
  return $\mathcal{O}_{Seg}$

MCPNet includes parallel Mamba–CNN paths and a bottom-up fusion process. For simplicity, the CNN branch and Mamba branch are referred to as the C-branch and M-branch, respectively. In the C-branch, we designed a detail preservation network (Section III-C) to extract rich detail features from large-size remote sensing images. Thanks to this design, additional Laplacian pyramid processing [15], [20] is avoided, saving memory and preserving more information. The output of the C-branch is a feature map $\mathbf{I}_{c1} \in \mathbb{R}^{(W/8)\times(H/8)\times 8C_1}$ and $\mathbf{I}_{c2} \in \mathbb{R}^{(W/16)\times(H/16)\times 16C_1}$, where W and H denote the width and height of the input image, respectively, and C_1 represents the number of basic channels in the C-branch (in the experiments, $C_1 = 32$).

The M-branch is built based on the VSS block [35] (Section III-C) and is primarily responsible for learning contextual information and modeling global semantics. The original image is downsampled by a factor of 4 before being fed into the M-branch. This operation reduces the computational burden while alleviating the local forgetting issue caused by Mamba's sequential scanning mechanism. Due to Mamba's efficient global modeling ability, we no longer require the ASPP module to expand the receptive field. The output of the M-branch is a feature map $\mathbf{I}_{m4} \in \mathbb{R}^{(W/128)\times(H/128)\times 8C_2}$, where C_2 represents the number of basic channels in the M-branch (in the experiments, $C_2 = 96$).

These three features will be input into the cascaded MG-FFM (Section III-D) for feature fusion and the gradual restoration of high-resolution features. Finally, an FCN head [59] is used to generate the semantic segmentation results. We present the training and inference processes of the algorithm using pseudocode, as illustrated in Algorithm 1, to further elaborate on the construction of MCPNet.

B. Dynamic Contextual Training Strategy

In FCtL [14], a stepwise training strategy is constructed to capture a broader range of contextual information. For a large-scale image, FCtL divides it into three different scales and then resizes the contextual images to a uniform size. A separate semantic segmentation network is trained for each scale, and finally, the outputs of these networks are fused to obtain the final semantic segmentation result. However, this approach is limited to three fixed context scales, making it susceptible to misleading information caused by object truncation. Additionally, the multistage, stepwise training strategy

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is highly complex and significantly slows down inference due to excessive slicing.

To address these issues, we propose a dynamic and continuous context training strategy, as shown in the upper left of Fig. 2. Given an original image $\mathbf{I} \in \mathbb{R}^{W\times H\times 3}$, in each training iteration, the entire image is rescaled before being fed into the network. Specifically, the scale range of the original image is $[\alpha^2 W \times H, \beta^2 W \times H]$, where α and β are scaling factors. Compared to FCtL, which divides each image into small 508 $\times$ 508 patches, our method adopts a larger sampling window of size $k \times k$ to improve data processing efficiency. The rescaled image is then randomly sampled using the sampling window, yielding an effective receptive field scale of $[(1/\beta_{\min}^2)k \times k, (1/\alpha_{\min}^2)k \times k]$ during training. When $(1/\gamma^2)k \times k = W \times H$, $\gamma \in [\alpha, \beta]$, the model attains global context information. As γ varies randomly, the model dynamically captures continuous contextual information throughout training. This strategy avoids the inefficiencies of multistep training while enhancing the model's adaptability to varying scales.

C. Parallel Heterogeneous Feature Extraction

1) Mamba Branch: The development of SSMs has demonstrated their potential as low-cost alternatives to Transformers. Traditional SSM, as linear time-invariant systems, are described by a state equation and an observation equation. The state equation includes the current state, inputs, and possible system parameters, describing how the system state evolves over time. Mathematically, this is expressed as follows:

$$h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t) \tag{1}$$

where $h(t) \in \mathbb{R}^N$ represents the system state at time step t, $h'(t)$ is the time derivative of the state vector at time t, and $x(t)$ is the input at time t. The state transition matrix is represented as $\mathbf{A} \in \mathbb{R}^{N\times N}$, and the input control matrix is represented as $\mathbf{B} \in \mathbb{R}^N$. The observation equation describes the relationship between the input and its state, mathematically expressed as

$$y(t) = \mathbf{C}h(t) + \mathbf{D}x(t) \tag{2}$$

where $y(t) \in \mathbb{R}$ represents the system output at time t, $\mathbf{C} \in \mathbb{R}^N$ is the output matrix, and $\mathbf{D} \in \mathbb{R}^1$ is used for residual connection operations. SSMs are continuous-time models, which pose challenges when integrated into deep learning networks. To address this, a time scale Δ is introduced, and a zero-order hold method is used to discretize the parameters $\mathbf{A}$ and $\mathbf{B}$ as follows:

$$\bar{A} = e^{\Delta\mathbf{A}}, \bar{B} = \left(\bar{A}\right)^{-1}\left(e^{\Delta\mathbf{A}} - \mathbf{I}\right) \cdot \Delta\mathbf{B}. \tag{3}$$

After discretization, the SSM equations are expressed as

$$h'(t) = \bar{A}h(t) + \bar{B}x(t) \tag{4}$$

$$y(t) = \mathbf{C}h(t) + \mathbf{D}x(t). \tag{5}$$

Mamba, as an extension of SSM, improves upon it by employing a selective scan mechanism that makes the parameters $\mathbf{B}$, $\mathbf{C}$, and Δ input-dependent, thereby enabling dynamic feature representation and enhancing its performance in handling complex tasks. In this study, we build our M-branch based on a deformed structure of Mamba, specifically the VSS model, within the realm of computer vision. For the downsampled input image $\mathbf{I}_{\mathrm{m}} \in \mathbb{R}^{(H/4)\times(W/4)\times 3}$, we first use patch embedding to partition it into small patches $\mathbf{I}_{\mathrm{embed}} \in \mathbb{R}^{(H/16)\times(W/16)\times C_2}$, which are then processed through stacked VSS blocks and downsample blocks to create hierarchical representation $\mathbf{I}_{\mathrm{mi}} \in \mathbb{R}^{(H/2^{2+i})\times(W/2^{2+i})\times 2^{i-1}C_2}$, where $i \in \{1, 2, 3, 4\}$. The specific structure of the VSS block is shown in Fig. 3(b). Each VSS block consists of two residual blocks. The first employs a 2-D selective scan (SS2D), which performs directional scanning from four orientations to achieve efficient global modeling. The second adopts a feedforward neural network (FFN) for nonlinear transformation and feature enhancement. Formally, the VSS block can be expressed as

$$\mathbf{I}_1 = \mathrm{Li} \circ \mathrm{LN} \circ \mathrm{SS2D} \circ \mathrm{SiLU} \circ \mathrm{DWC} \circ \mathrm{Li} \circ \mathrm{LN}(\mathbf{I}_{\mathrm{in}}) \tag{6}$$

$$\mathbf{I}_2 = \mathbf{I}_1 + \mathbf{I}_{\mathrm{in}} \tag{7}$$

$$\mathbf{I}_{\mathrm{out}} = \mathbf{I}_2 + \mathrm{FFN}(\mathrm{LN}(\mathbf{I}_2)) \tag{8}$$

where LN, SS2D, SiLU, DWC, and Li denote layer normalization, SS2D, the SiLU activation function, depthwise separable convolution, and linear layer, respectively. $\mathbf{I}_{\mathrm{in}}, \mathbf{I}_1, \mathbf{I}_2, \mathbf{I}_{\mathrm{out}} \in \mathbb{R}^{W_v\times H_v\times C_v}$ denote the input feature map, the output feature map from the SS2D block, the output feature map from the first residual structure, and the output feature map from the VSS block, respectively; W_v and H_v denote the width and height of the feature map, respectively; $\circ$ denotes function composition; and FFN denotes the feedforward neural network.

Then, these $\mathbf{I}_{\mathrm{m4}} \in \mathbb{R}^{(H/128)\times(W/128)\times 8C_2}$ are sent to the cascaded MG-FFM for feature fusion with the C-branch features and provide rich category semantic guidance during both the feature fusion and decoding processes.

2) Detail Preservation Network: Existing research has shown that different convolution kernel sizes [60], [61], [62], [63], [64] and different downsampling strategies [65], [66], [67], [68] have led to beneficial improvements in image classification, semantic segmentation, and detection tasks. Inspired by these works, we constructed a CNN-based detail preservation network for the C-branch of MCPNet. The specific structure of this network is shown in Fig. 3(a).

The C-branch feature extraction comprises two stages. In the first stage, a tree-like network composed of multiple downsampling branches processes the input feature map $\mathbf{I}_{\mathrm{cin}} \in \mathbb{R}^{W\times H\times C}$ (with $C = 16$ in experiments) to preserve different detail features. The process represented by each parallel branch is given as follows:

$$\begin{aligned} \mathbf{I}_{\mathrm{C3}} &= \mathrm{Conv}_{3\times 3}(\mathbf{I}_{\mathrm{cin}}) \\ \mathbf{I}_{\mathrm{F}} &= \mathrm{Conv}_{1\times 1}(\mathrm{Focus}(\mathbf{I}_{\mathrm{cin}})) \\ \mathbf{I}_{\max} &= \mathrm{Maxpool}(\mathbf{I}_{\mathrm{cin}}) \end{aligned} \tag{9}$$

where $\mathrm{Conv}_{3\times 3}$ (stride = 2 and kernel = 3), $\mathrm{Conv}_{1\times 1}$ (stride = 1 and kernel = 1), and Maxpool (stride = 2 and kernel = 2) are used for downsampling and feature transformation. The Focus module halves the spatial size and quadruples the channel count by spatial rearrangement and concatenation.

The outputs are then fused via channel-wise concatenation and a 1 $\times$ 1 convolution to integrate global channel

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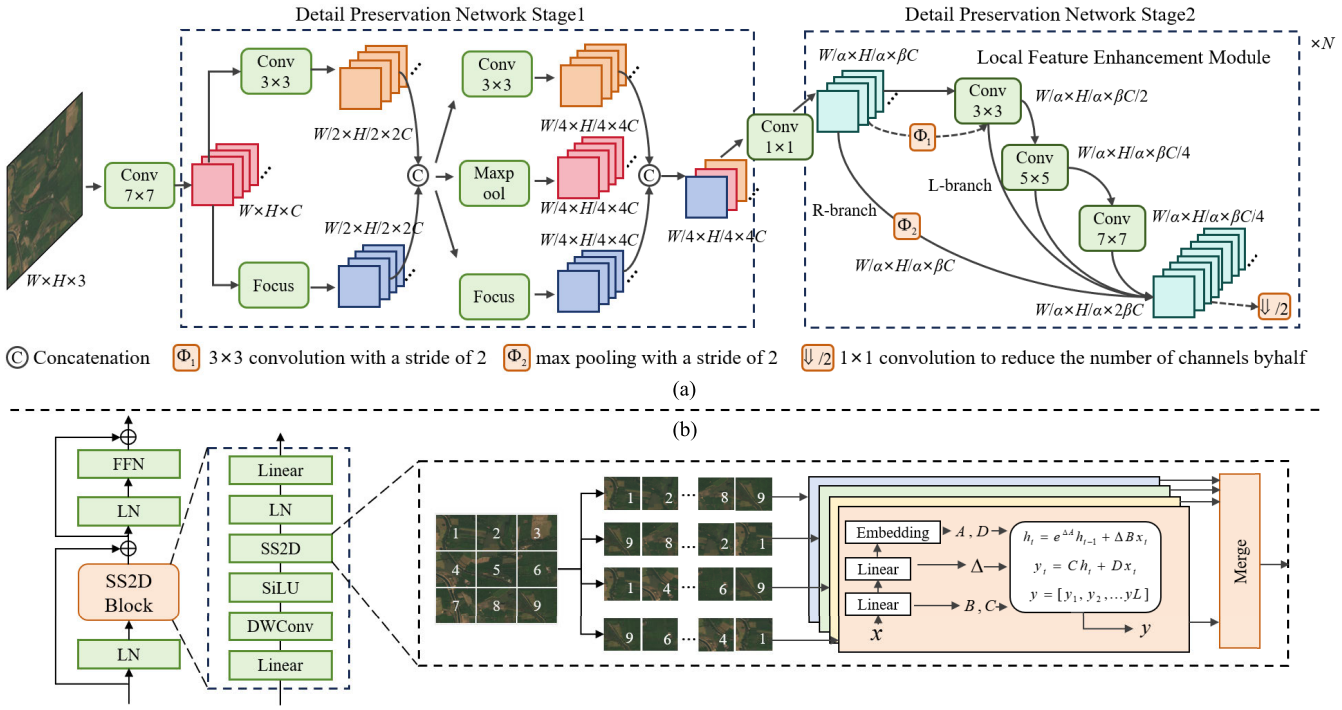


Fig. 3. Pipeline of parallel heterogeneous feature extraction. (a) Structure of detail preservation network. In this process, the image first undergoes a 7 × 7 convolution, followed by parallel multipath downsampling. After each path, a 1 × 1 convolution is applied to unify the branch channel dimensions, and features are concatenated along the channel dimension. Subsequently, the features are sent into multiple stacked LFEMs, where different-sized convolutional kernels are used to capture local features with varying receptive fields. These features are then preserved with fine-grained details through dense connections. (b) Structure of the VSS block and the SS2D process. SS2D scans the image blocks in four directions, and the results are aggregated to form the output. In this figure, N refers to the number of stacked LFEMs, and α and β are scaling factors. In the experiments, $\alpha = 8$ and 16 and $\beta = 8$ and 16.

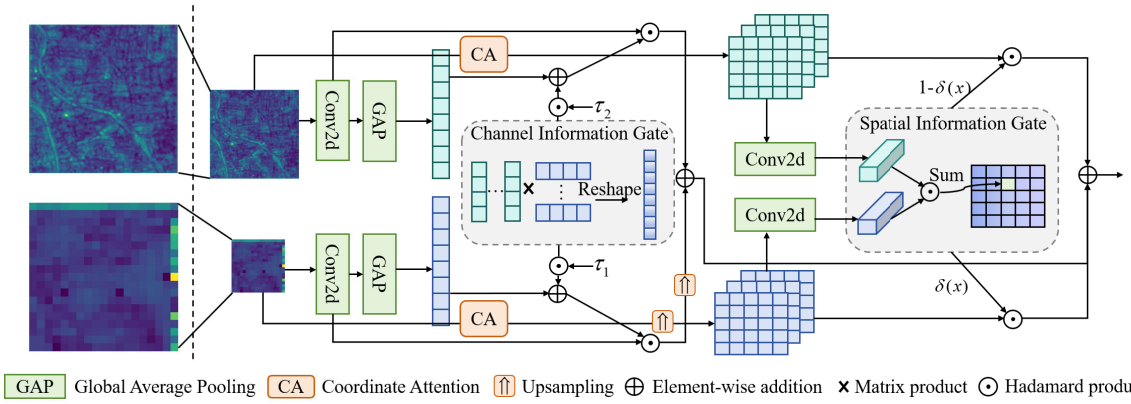


Fig. 4. Detailed architecture of the MG-FFM. The left side of the vertical dashed line shows the visualization results of the feature maps from different branches, while the right side illustrates the specific process of feature fusion. The two dashed boxes highlight the construction methods of the channel and spatial information gates.

information, yielding intermediate features $\mathbf{I}_{\text{dp1}} \in \mathbb{R}^{(W/2)\times(H/2)\times 2C}$ and $\mathbf{I}_{\text{dp2}} \in \mathbb{R}^{(W/4)\times(H/4)\times 4C}$ that were given as follows:

$$\mathbf{I}_{\text{dp1}} = \text{Conv}_{1\times1}(\mathbb{C}(\mathbf{I}_{\text{C3}}, \mathbf{I}_{\text{F}})) \tag{10}$$

$$\mathbf{I}_{\text{dp2}} = \text{Conv}_{1\times1}(\mathbb{C}(\mathbf{I}_{\text{C3}}, \mathbf{I}_{\text{F}}, \mathbf{I}_{\text{max}})) \tag{11}$$

where $\mathbb{C}$ denotes the concatenation of feature maps along the channel dimension.

In the second stage, $\mathbf{I}_{\text{dp2}}$ is projected to $\mathbf{I}_{\text{dp3}} \in \mathbb{R}^{(W/4)\times(H/4)\times 8C}$ via a 1 × 1 convolution and then passed through $N \in \{1, 2, 3, 4\}$ stacked local feature enhancement modules (LFEMs). Each LFEM consists of two branches. The L-branch applies a series of convolutions of sizes 3 × 3, 5 × 5, and 7 × 7 with a stride of 1. The 3 × 3 convolution reduces channels by half, while the others retain channel dimensions. Each convolution is followed by a batch normalization and ReLU activation. The R-branch generates features $\mathbf{I}_{\text{R3}}$, and all outputs are concatenated and fused as follows:

$$\mathbf{I}_{\text{dpout}} = \text{Conv}_{1\times1}(\mathbb{C}(\mathbf{I}_{\text{R3}}, \mathbf{I}_{\text{c3}}, \mathbf{I}_{\text{c5}}, \mathbf{I}_{\text{c7}})). \tag{12}$$

This design enhances local multiscale context while preserving spatial resolution due to the stride-1 convolutions. Moreover, for $N = 1$ and 3, both L- and R-branches downsample the feature maps using 3 × 3 convolutions and max

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pooling with stride 2 [denoted as Φ_1 and Φ_2 in Fig. 3(a)]. The resulting features are then compressed via a 1×1 convolution to halve the channel count [see $\Downarrow /2$ in Fig. 3(a)].

D. Multigated Feature Fusion Module

With the increase in input image size, the model attains a larger receptive field while simultaneously incorporating more information, which may contain additional redundancy and noise. Furthermore, as illustrated on the left of Fig. 4, the features extracted by the M-branch and C-branch exhibit distinct characteristic trends. Specifically, the C-branch primarily focuses on capturing high-frequency details, whereas the M-branch demonstrates a stronger capability for abstract semantic modeling. Moreover, these two branches exhibit significant discrepancies in their feature spaces concerning spatial resolution, spatial dimensionality, and channel activation responses.

To effectively integrate heterogeneous features from the C-branch and M-branch, which are responsible for fine-grained details and high-level semantics, respectively, we propose the MG-FFM, as shown on the right of Fig. 4.

Given feature maps $\mathbf{F}_{\text{sp}}$ and $\mathbf{F}_{\text{co}}$ from the C-branch and M-branch, we first align their channels via 1×1 convolutions followed by channel attention [69], producing $\mathbf{f}_{\text{sp}}, \mathbf{f}_{\text{co}} \in \mathbb{R}^{H' \times W' \times C'}$. We then construct the corresponding multihead channel descriptors $\Gamma_{\text{sp}}, \Gamma_{\text{co}} \in \mathbb{R}^{(C'/\gamma) \times \gamma}$ to encode global semantics

$$\Gamma_{\text{sp}} = \text{MLP}\big(\text{GAP}\big(\text{Conv}_{1\times1}\big(\mathbf{F}_{\text{sp}}\big)\big)\big) \tag{13}$$

$$\Gamma_{\text{co}} = \text{MLP}(\text{GAP}(\text{Conv}_{1\times1}(\mathbf{F}_{\text{co}}))) \tag{14}$$

where GAP denotes the global average pooling, MLP represents the multilayer perceptron (MLP). Next, an affinity matrix $\mathfrak{R} \in \mathbb{R}^{(C'/\gamma)\times(C'/\gamma)}$ is constructed to capture cross-modal semantic correlations

$$\mathfrak{R} = \mathbf{F}_{\text{sp}} \mathbf{F}_{\text{co}}{}^{\top}. \tag{15}$$

To refine channel interactions, we introduce two learnable residual gates τ_1 and τ_2 and obtain final channel gating vectors

$$\mathbf{G}_{\text{co}} = \delta(\Gamma_{\text{co}} + \tau_1(\text{MLP}(\mathfrak{R}))) \tag{16}$$

$$\mathbf{G}_{\text{sp}} = \delta\big(\Gamma_{\text{sp}} + \tau_2(\text{MLP}(\mathfrak{R}))\big) \tag{17}$$

where δ denotes the sigmoid function.

To emphasize salient spatial regions, we apply coordinate attention [70] to both $\mathbf{F}_{\text{sp}}$ and $\mathbf{F}_{\text{co}}$, obtaining enhanced features $\bar{F}_{\text{sp}}$ and $\bar{F}_{\text{co}}$, respectively. Notably, $\mathbf{F}_{\text{co}}$ is first processed by coordinate attention and then upsampled to match the resolution. Next, we extract position-wise feature vectors $\overrightarrow{v}_{sp}$ and $\overrightarrow{v}_{co}$ from the enhanced maps. The spatial affinity gate is then computed as follows:

$$\bar{G}_{\text{pos}} = \delta\big(\epsilon\big(\big(\text{MLP}\big(\overrightarrow{v}_{sp}\big) \odot \text{MLP}\big(\overrightarrow{v}_{co}\big)\big)\big)\big) \tag{18}$$

where $\odot$ denotes the Hadamard product and ϵ aggregates across channels. This gate adaptively weighs contributions; when $\bar{G}_{\text{pos}}$ is high, M-branch features dominate; otherwise, C-branch details are emphasized. The final fused feature representation $\mathbf{F}_{\text{fusion}}$ is given as follows:

$$\mathbf{F}_{\text{fusion}} = \mathbf{G}_{\text{sp}} \odot \mathbf{F}_{\text{sp}} + \text{Up}(\mathbf{G}_{\text{co}} \odot \mathbf{F}_{\text{co}}) + \bar{G}_{\text{pos}} \odot \overrightarrow{v}_{co} + \big(1 - \bar{G}_{\text{pos}}\big) \odot \overrightarrow{v}_{sp}. \tag{19}$$

This fusion strategy enables progressive integration of semantic abstraction and spatial details, enhancing both feature complementarity and robustness by filtering redundancy across branches.

E. Optimization

During the training process, we employ a joint loss to constrain the model optimization. The final semantic segmentation head($\mathcal{L}_{\text{Seg}}$), along with the M-branch auxiliary head($\mathcal{L}_{\text{AuxVss}}$) and the decoder auxiliary head($\mathcal{L}_{\text{AuxDecoder}}$) in deep supervision learning, all utilize multiclass CE loss. The output $\mathbf{I}_{\text{m4}}$ from the M-branch is also fed into a super-resolution reconstruction head (SR Head) [15], which consists of three convolutional layers and three upsampling operations for feature reconstruction to preserve the spatial structure of M-branch. The reconstruction head simultaneously generates both the reconstructed image $\mathbf{I}_{\text{img}}$ and the reconstructed features $\mathbf{I}_{\text{re}}$. Then, an mse loss is used as the reconstruction loss $\mathcal{L}_{\text{re}}$

$$\mathcal{L}_{\text{re}} = \big\|\mathbf{I} - \text{Up}\big(\mathbf{I}_{\text{img}}\big)\big\|_2^2 \tag{20}$$

where Up and $\mathbf{I}$ denote the upsampling operation and the input image, respectively. A feature affinity loss $\mathcal{L}_{\text{aff}}$ [71] is used to ensure the stability of the reconstruction process

$$\mathcal{L}_{\text{aff}} = \big\|\mathbf{I}_{\text{m4}}^{\top}\mathbf{I}_{\text{m4}} - \text{Down}(\mathbf{I}_{\text{re}})^{\top}\text{Down}(\mathbf{I}_{\text{re}})\big\| \tag{21}$$

where Down denotes the downsampling operation. The total loss function is given by

$$\mathcal{L}_{\text{all}} = \mathcal{L}_{\text{seg}} + w_1\mathcal{L}_{\text{AuxVss}} + w_2\sum_{i=1}^{2}\mathcal{L}_{\text{AuxDecoderi}} + w_3\mathcal{L}_{\text{re}} + w_4\mathcal{L}_{\text{aff}} \tag{22}$$

where w_1–w_4 are set as 1, 0.4, 0.5, and 1, respectively. We optimize the entire objective function in an end-to-end manner, minimizing it through the whole process.

IV. Experiments and Analysis

We designed experiments on three datasets to verify the effectiveness of MCPNet. First, we introduce the basic information of the three datasets: the evaluation metrics, the experimental environment, and the training details. Then, Sections IV-C–IV-E describe the experimental design and results on the DeepGlobe, our self-constructed Yaan datasets, and Inria Aerial, respectively. Finally, a comprehensive ablation study is conducted to verify the effectiveness of each network component (Section IV-F).

A. Datasets and Evaluation Metrics

To evaluate the performance of the proposed method, we conducted experiments on the self-constructed Yaan dataset and two widely used large-size remote sensing image datasets: DeepGlobe [30] and Inria Aerial [31]. A summary of these datasets is presented in Table I, including image size, GSD, the land-cover classes contained in each dataset, the per-class pixel distribution for each dataset, and the dataset partitioning scheme. Some data examples of them are shown in Fig. 5.

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1) DeepGlobe Dataset: This dataset consists of 803 large-size remote sensing images, which were randomly divided into training, validation, and test sets, containing 455, 207, and 142 images, respectively. This dataset includes seven classes of landscape regions, with each image having an image size of 2448 × 2448 pixels. Following the practice in [13], the "unknown" class is excluded when computing quantitative evaluation metrics.

2) Inria Aerial Dataset: This dataset comprises 180 large-size remote sensing images, each with an image size of 5000 × 5000 pixels. The dataset is annotated with binary masks distinguishing buildings and nonbuildings. We randomly split the dataset into training, validation, and test sets, consisting of 126, 27, and 27 images, respectively.

3) Yaan Dataset: For the collected UAV images of Yaan, we generated a comprehensive orthophotograph of the study area using aerial triangulation combined with dense point cloud reconstruction and digital surface model (DSM) generation. This process ensures annotation continuity and minimizes data redundancy, thereby facilitating the learning of more generalizable features. The resulting orthophotograph has a GSD of 0.03 m. All preprocessing steps, including orthorectification and image mosaicking, were performed using Agisoft PhotoScan. Subsequently, various landscape objects were annotated seamlessly using the polygon creation tool in ArcGIS. The annotations were independently carried out by two expert annotators to ensure label accuracy and consistency. The class-wise distribution of land-cover types in the dataset is presented in Table I.

During model training and testing, the study area was partitioned into 70 nonoverlapping images, each with a resolution of 6000 × 4000 pixels. This dataset encompasses seven landscape classes and was randomly divided into training, validation, and test subsets containing 40, 10, and 20 images, respectively.

4) Evaluation Metrics: We report intersection over union (IoU) and $F1$-score for each class, as well as their mean values (mean intersection over union (mIoU) and mF1). In addition, we measure GPU memory usage (MB) and inference speed (FPS).

B. Experimental Setup

The experiments in this study were implemented based on the MMSegmentation codebase [78]. All training processes were conducted on a single RTX 4090 GPU. The model parameters were optimized using the stochastic gradient descent (SGD) algorithm. Due to hardware resource limitations, we set the batch sizes for the DeepGlobe, Inria Aerial, and Yaan datasets to 8, 2, and 2, respectively. The initial learning rates were set to 10^{-2}, 5×10^{-3}, and 10^{-2}. A polynomial decay strategy with a decay factor of 0.9 was applied during training. The total number of training iterations was set to 80k for DeepGlobe, 160k for Inria Aerial, and 40k for the Yaan dataset. Additionally, we utilized the "gpustat" tool to monitor GPU memory usage during model inference.

C. Evaluation on the DeepGlobe Dataset

1) Memory Consumption Analysis of Conventional Semantic Segmentation Methods: In this section, we first analyze the GPU memory consumption of several leading and representative conventional semantic segmentation models with respect to varying input image sizes, including convolution-based models (UNet [5], PSPNet [75], and DeepLabv3+ [73]), attention-based models (DANet [74], CCNet [77], and EMCAD [76]), Transformer-based models (Segmenter [22], SegFormer [6], and Swin-UNet [21]), Mamba-based models (RS-Mamba [7] and VM-UNet [72]), and hybrid models (CMTFNet [9], RS3Mamba [8], and CTCFNet [29]). During inference, the batch size is set to 1, and eight different input sizes ranging from 512 to 5000 pixels are tested. Due to structural and implementation constraints of certain models (e.g., multiscale connections and predefined resolution requirements that prevent processing arbitrary image sizes), slight modifications were made to some resolution settings. Specifically, the tested input sizes include [512, 768, 1024, 1536, 2048, 2448 (2432), 4096, 5000]. Swin-UNet and CTCFNet utilize input sizes of [448, 672, 896, 1344, 1792, 2464, 3584, 4928]. When processing images exceeding 6.25 million pixels, our method employs a large-patch slicing window (2000 × 2000) with a stride of 1000 × 1000, while other resolutions are inferred using full-sized images.

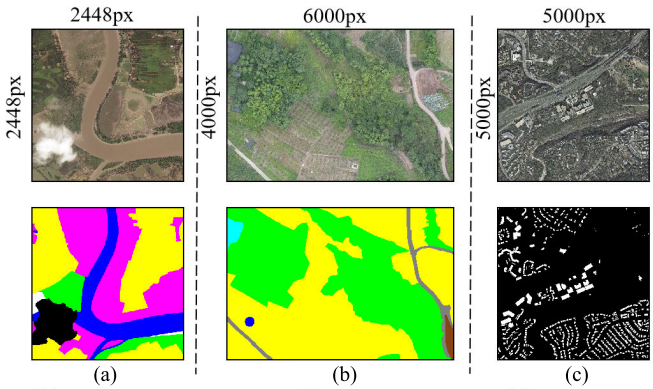


Fig. 5. Examples from the DeepGlobe, Yaan, and Inria Aerial datasets. (a) DeepGlobe dataset consists of seven land-cover categories, with each image having an image size of 2448 × 2448. (b) Yaan dataset contains seven land-cover categories, with each image having an image size of 6000 × 4000. (c) Inria Aerial dataset includes buildings and background, with each image having an image size of 5000 × 5000.

As shown in Table II, conventional methods exhibit significant memory growth as input size increases due to the lack of memory-efficient designs. Transformer-based models (e.g., SegFormer and Segmenter) require computing global self-attention, which involves pairwise similarity calculations across all pixels, leading to a sharp increase in memory consumption and limiting their applicability to large-size images. Similarly, DANet incorporates global attention mechanisms similar to Transformers and also experiences a steep rise in memory usage as the input size increases. UNet-style architectures (e.g., UNet, Swin-UNet, RS-Mamba, VM-UNet, EMCAD, and CMTFNet) retain multiple intermediate feature maps across different resolutions, preventing them from efficiently discarding previous-stage features during inference to reduce memory consumption. As a result, these methods are not well-suited for handling large-size images. In contrast, our method efficiently extracts features by designing

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TABLE I
OVERVIEW OF THE THREE DATASETS

DataSet	Image Size (px)	GSD (m)	Category & pixel ratio	Train	Val	Test
DeepGlobe [30]	2448 × 2448	0.5	urban (10.80%), agriculture (57.74%), rangeland (8.48%), forest (11.16%), water (3.31%), barren (8.45%), unknown (0.05%)	455	207	142
Yaan Dataset	6000 × 4000	0.03	built-up land (7.33%), farmland (27.18%), wasteland (5.71%), forest (39.95%), water (1.12%), road (3.88%), background (14.83%)	40	10	20
Inria Aerial [31]	5000 × 5000	0.3	building (15.78%), background (84.22%)	126	27	27

TABLE II
GPU MEMORY CONSUMPTION DURING INFERENCE (BATCH SIZE = 1) FOR CONVENTIONAL SEMANTIC SEGMENTATION METHODS. "∗" INDICATES THAT A SIZE OF 2432 WAS USED INSTEAD OF 2448 DURING INFERENCE

Method	512 × 512	768 × 768	1024 × 1024	1536 × 1536	2048 × 2048	2448 × 2448	4096 × 4096	5000 × 5000
Segmenter [22]	**576**	**684**	**980**	3566	12994	19586	OOM	OOM
SegFormer [6]	780	1142	1510	4056	10356	OOM	OOM	OOM
Swin-UNet [21]	876	1232	1696	3048	4944	6006	12014	22782
CMTFNet* [9]	980	1240	1780	3120	5188	6706	18354	23234
VM-UNet* [72]	984	1388	1898	3376	5660	7714	19820	OOM
DeepLabv3+ [73]	990	1494	2248	3770	6140	8476	22410	OOM
DANet [74]	998	1854	3624	12958	OOM	OOM	OOM	OOM
PSPNet [75]	1070	1494	2040	3840	6170	8458	20618	24018
EMCAD* [76]	1076	1492	2214	4104	6778	9282	23482	OOM
CCNet [77]	1096	1370	1772	3652	5868	8048	21084	OOM
RS3Mamba [8]	1108	1678	2446	4624	7690	10460	21604	OOM
RS-Mamba* [7]	1172	1796	2628	4906	8182	11260	OOM	OOM
UNet [5]	1342	2064	3652	7400	12328	17790	OOM	OOM
CTCFNet [29]	1394	1914	2920	5414	9524	22118	OOM	OOM
MCPNet(Ours)	806	914	1126	**1586**	**2260**	**2794**	**3120**	**3526**

two asymmetric heterogeneous lightweight branches while avoiding excessive preservation of multilevel features, thereby significantly reducing memory consumption.

2) Quantitative Comparison With Conventional Semantic Segmentation Methods: Next, we apply MCPNet to the DeepGlobe dataset and compare its performance with the conventional semantic segmentation methods listed in Table II. Due to the high memory consumption of these methods, we apply slice-based training to them. Specifically, the input images are first zero-padded to 2560 × 2560 and then cropped into 512 × 512 patches without overlap for training and testing. Given the specific characteristics of certain models, images for Swin-UNet and CTCFNet are cropped into 448 × 448 patches, while our method takes 2448 × 2448 as the input size.

As shown in Table III, we observe that pure Transformer-, Mamba-, and CNN-based models exhibit limited performance under these training conditions, potentially due to an imbalance between global and local modeling. Furthermore, pure Transformer and Mamba models suffer from high training complexity and a lack of spatial prior knowledge, constraining their overall performance. In contrast, attention-based and hybrid models demonstrate notable performance advantages. However, conventional methods still show significant performance differences from our approach due to the loss of contextual information and geometric discontinuities caused by patch-wise processing. Our method achieves an mIoU improvement of 0.73%–28.99% and an mF1-score improvement of 0.82%–27.67% over the compared methods.

3) Quantitative Comparison With Memory-Efficient Methods: Furthermore, we compare our method with several memory-efficient semantic segmentation approaches, including UNetFormer [10], BiSeNetv1 [12], CM-UNet [11], CascadePSP [16], PointRend [79], MagNet [17], PPN [18], GLNet [13], ISDNet [15], FCtL [14], and WSDNet [20]. For these methods, we use the original images with a size of 2448 × 2448 for both training and testing. Due to implementation constraints in the model code, the input sizes for UNetFormer and CM-UNet are slightly resized to 2432 × 2432. As presented in Table III, our method achieves an mIoU improvement of 2.0%–12.43% and an mF1-score improvement of 0.66%–9.77%.

Furthermore, in terms of class-wise semantic segmentation accuracy, MCPNet outperforms all comparison methods in five categories: "agriculture," "rangeland," "forest," "water," and "barren." This result fully demonstrates the effectiveness of our method in leveraging rich global semantic information from large-size images. However, for the "urban" category, conventional patch-based semantic segmentation methods exhibit clear performance advantages. This phenomenon may be attributed to the fact that certain buildings in the DeepGlobe dataset occupy only a small proportion of the entire image. When directly processing large-scale images, fine-grained architectural details are likely to be overwhelmed by broader semantic regions, leading to a reduction in semantic segmentation accuracy. In terms of inference speed, aside from GLNet and FCtL—whose performance suffers significantly due to the excessive number of patches—other memory-efficient methods achieve substantially faster inference than

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TABLE III
QUANTITATIVE COMPARISON ON THE DEEPGLOBE DATASET. "†" DENOTES THE RESULTS COLLECTED FROM [20]

Method	IoU for Each Class(%)						mIoU(%)	mF1(%)	Memory(MB)	FPS
	Agriculture	Rangeland	Urban	Forest	Water	Barren				
Conventional sematic segmentation method										
Swin-UNet [21]	68.05	0.00	81.91	64.17	45.61	22.94	47.11	58.19	876	1.31
RS-mamba [7]	76.37	0.51	83.48	74.22	66.18	44.10	57.48	67.64	1172	1.03
VM-UNet [72]	76.60	4.76	82.67	72.08	68.50	42.00	57.77	68.44	984	1.08
Segmenter [22]	80.70	31.99	82.30	77.68	73.10	52.05	66.30	78.07	**576**	2.10
SegFormer [6]	83.65	40.52	87.53	78.48	78.88	58.32	71.23	81.99	780	1.58
UNet [5]	84.88	42.16	86.74	78.59	77.07	61.84	71.88	82.59	1342	2.71
RS3Mamba [8]	84.09	38.31	87.37	79.73	81.34	60.84	71.95	82.34	1108	1.13
EMCAD [76]	84.16	41.68	87.25	80.21	82.56	60.15	72.67	83.00	1076	1.94
CTCFNet [29]	84.53	42.20	**92.12**	79.63	80.59	60.47	73.26	83.36	1394	0.71
PSPNet [75]	86.84	46.24	87.19	81.12	82.68	64.71	74.79	84.68	1070	2.78
DeepLabv3+ [73]	85.88	46.78	87.66	82.05	84.05	63.18	74.93	84.75	990	2.63
CCNet [77]	85.89	45.43	88.23	80.72	85.08	64.59	74.99	84.73	1096	2.15
CMTFNet [9]	85.96	45.32	87.84	82.37	83.74	65.25	75.08	84.81	980	1.74
DANet [74]	86.48	46.62	87.98	81.41	84.48	65.25	75.37	85.04	998	2.47
Memory-Efficient Methods										
UNetFormer [10]	81.28	29.11	73.21	74.63	72.92	50.83	63.67	76.09	2238	11.77
BiseNetv1 [12]	81.13	35.53	72.46	69.98	74.34	56.98	65.08	77.71	1996	10.64
CM-UNet [11]	82.64	31.93	73.30	75.24	80.44	54.39	66.33	78.17	2168	6.06
CascadePSP †[16]	/	/	/	/	/	/	68.50	79.70	3236	/
PointRend [79]	86.10	43.95	75.43	82.10	85.33	62.60	72.59	83.48	1593	10.33
MagNet-fast †[17]	/	/	/	/	/	/	71.80	/	1559	/
PPN †[18]	/	/	/	/	/	/	71.90	/	1193	/
GLNet [13]	86.13	42.72	75.33	82.07	82.35	65.07	72.28	82.94	2780	0.12
MagNet †[17]	/	/	/	/	/	/	72.90	/	1559	/
ISDNet [15]	86.53	43.45	76.54	82.33	84.16	67.09	73.35	83.67	2154	**14.19**
FCtL [14]	86.54	45.29	76.42	82.39	84.58	65.76	73.50	83.85	5682	0.03
WSDNet †[20]	/	/	/	/	/	/	74.10	85.20	1876	/
MCPNet(Ours)	**87.64**	**53.01**	75.50	**85.24**	**85.13**	**70.10**	**76.10**	**85.86**	2794	9.20

conventional patch-based methods, further highlighting the computational efficiency advantage of processing large-size images directly.

4) Memory–Accuracy Tradeoff: To further explore the tradeoff between semantic segmentation accuracy and memory consumption, we designed a comparative experiment as illustrated in Table IV. We selected DANet, CCNet, and CMTFNet—three of the best-performing conventional patch-based methods—for further investigation of their potential on large-scale image semantic segmentation. Meanwhile, we also included several memory-efficient models (CM-UNet, UNetFormer, and BiSeNetv1) as well as our proposed MCPNet for a comprehensive comparison.

Several noteworthy observations emerged from this comparison. Specifically, most comparison methods exhibit a "rise-then-drop" trend in semantic segmentation accuracy as the input size increases. Initially, enlarging the input size helps models capture more complete object shapes and broader contextual semantics, thereby improving semantic segmentation accuracy. However, further increases in size introduce excessive redundant and interference information, which in turn limits or even degrades model performance. In contrast, MCPNet adopts a dynamic contextual training strategy and an asymmetric hybrid structure, enabling it to effectively leverage rich contextual cues while filtering out redundancy through the MG-FFM module. As a result, MCPNet achieves high accuracy with balanced memory consumption.

Conventional models such as CCNet and DANet, which combine convolutional layers with global attention mechanisms, perform well at smaller input sizes. However, the sharp increase in memory usage and the resulting performance degradation with larger inputs significantly limit their applicability to large-scale remote sensing imagery.

To provide a more intuitive analysis of performance trends across different input sizes, we plotted the IoU of each model for each category as line charts in Fig. 6. The analysis reveals the following. As the input image size increases, for the "urban" category, all models exhibit similar patterns—IoU first increases and then decreases, peaking at the 768 × 768 input size. This aligns with the observation in Table III that patch-based methods achieve better performance on the "urban" class. For the other five categories, comparison methods (except ours) show a similar trend, with varying degrees of fluctuation across categories. In contrast, MCPNet demonstrates a consistent upward trend in these categories, further confirming its robustness and adaptability for large-size remote sensing image semantic segmentation.

5) Visualization Analysis of Inference Results for Different Models: The qualitative analysis on the DeepGlobe dataset is shown in Fig. 7. The third to sixth columns of the visualization represent stitched outputs from conventional patch-based methods, while the 7th–12th columns show global inference results. Evidently, conventional methods exhibit noticeable tiling artifacts when reconstructing

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TABLE IV
MODEL PERFORMANCE AND MEMORY CONSUMPTION WITH VARYING INPUT SIZES

Method	Metric	input size(px) 256 × 256	512 × 512	768 × 768	1224 × 1224	2448 × 2448
CM-UNet[11]	**mIoU(%)**	68.41	72.84	72.72	65.00	66.33
	mF1(%)	79.56	83.13	82.59	76.90	78.17
	Tr.(bs=2) / Inf.(bs=1) Mem.(MB)	1269 / 600	2049 / 632	3545 / 742	7593 / 1020	20455 / 2168
UNetFormer[10]	**mIoU(%)**	69.42	70.49	71.56	65.92	63.57
	mF1(%)	80.29	82.14	81.69	77.57	76.09
	Tr.(bs=2) / Inf.(bs=1) Mem.(MB)	1047 / 588	1485 / 608	2495 / 708	5563 / 1006	18063 / 2238
BiSeNetv1[12]	**mIoU(%)**	70.79	71.83	73.75	68.99	65.08
	mF1(%)	81.67	82.34	83.77	80.47	77.71
	Tr.(bs=2) / Inf.(bs=1) Mem.(MB)	915 / 576	1411 / 614	2085 / 716	4869 / 909	15909 / **1996**
DANet [74]	**mIoU(%)**	73.43	75.37	74.44		
	mF1(%)	83.54	85.04	84.11	OOM	OOM
	Tr.(bs=2) / Inf.(bs=1) Mem.(MB)	1981 / 909	4281 / 998	10137 / 1854		
CCNet [77]	**mIoU(%)**	73.37	74.99	76.03	68.11	
	mF1(%)	83.55	84.73	85.10	79.48	OOM
	Tr.(bs=2) / Inf.(bs=1) Mem.(MB)	1863 / 923	4189 / 1096	7979 / 1370	17791 / 2615	
CMTFNet [9]	**mIoU(%)**	70.64	75.08	69.86	59.39	
	mF1(%)	81.33	84.81	80.79	72.59	OOM
	Tr.(bs=2) / Inf.(bs=1) Mem.(MB)	1767 / 891	3961 / 980	8377 / 1240	19245 / 2263	
MCPNet(Ours)	**mIoU(%)**	65.94	71.32	73.59	73.93	**76.10**
	mF1(%)	76.93	82.24	83.41	84.34	**85.86**
	Tr.(bs=2) / Inf.(bs=1) Mem.(MB)	1407 / 736	1609 / 806	2133 / 914	3467 / 1370	**7656** / 2794

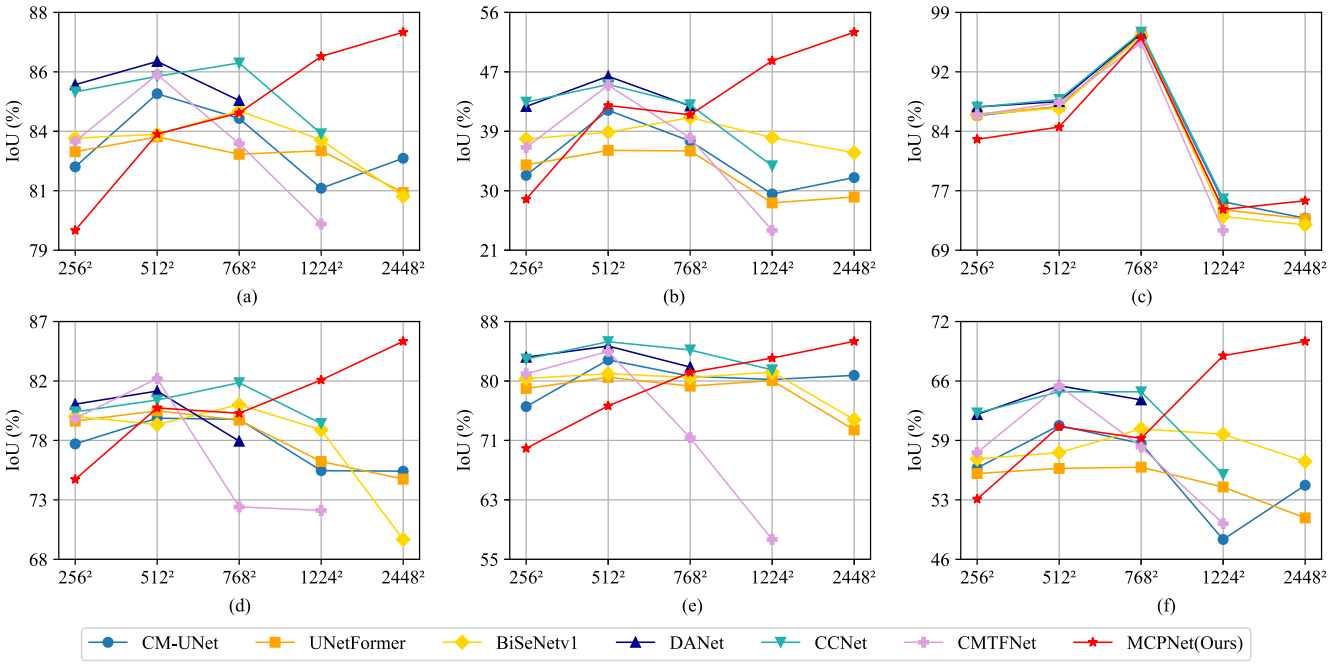


Fig. 6. Class-wise IoU variations on the DeepGlobe dataset under different input image sizes. (a) Agriculture. (b) Rangeland. (c) Urban. (d) Forest. (e) Water. (f) Barren.

segmented patches, whereas our approach effectively mitigates such artifacts.

The first four rows display the model's fine-grained pixel classification capabilities. In the first and second scenes of the first and second rows, our method identifies the most complete and continuous rivers, while in the first-row scene, our method also accurately identifies the overall shape of the agricultural land. In the second-row scene, our method successfully detects rangeland areas within large agricultural zones, while the results of other methods are more fragmented. In this dataset, the "rangeland" and "agriculture" classes have similar color and texture characteristics, but "agriculture" has stronger structural features. The ability to extract and leverage this feature is crucial for achieving accurate semantic segmentation. The third and fourth rows illustrate the model's performance in urban regions. Compared to other methods, our approach more accurately delineates the structural shapes of urban areas. Although there are occasional

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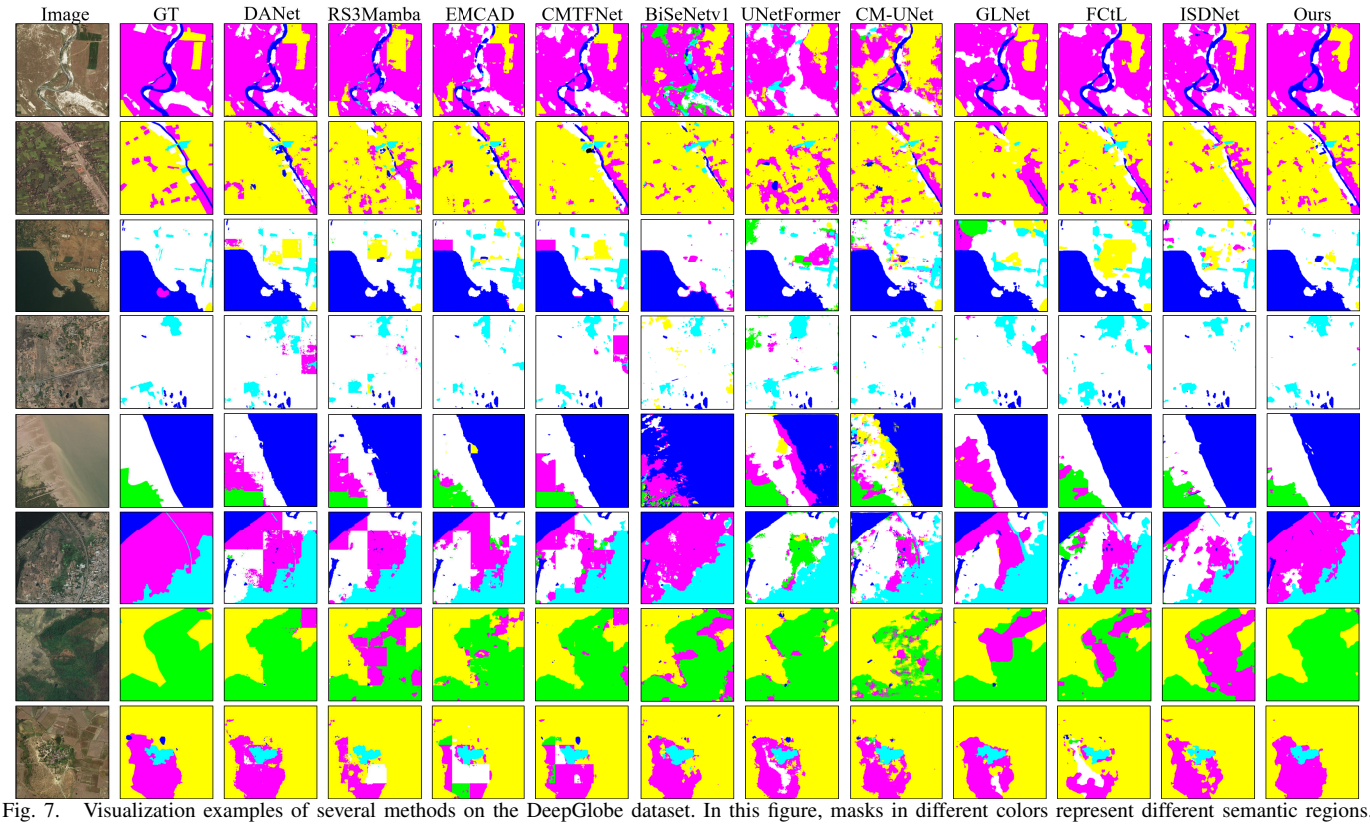


Fig. 7. Visualization examples of several methods on the DeepGlobe dataset. In this figure, masks in different colors represent different semantic regions. Specifically, cyan represents "urban," yellow represents "agriculture," purple represents "rangeland," green represents "forest," blue represents "water," white represents "barren," and black represents "unknown."

TABLE V

QUANTITATIVE COMPARISON ON THE YAAN DATASET. "WASTEL.," "FARM.," "BUILT-UP.," AND "BACKG." DENOTE "WASTELAND," "FARMLAND," "BUILT-UP LAND," AND "BACKGROUND," RESPECTIVELY

Method	IoU for Each Class(%)							mIoU(%)	mF1(%)	Memory(MB)	FPS
	Wastel.	Forest	Farm.	Water	Road	Built-up.	Backg.				
Conventional sematic segmentation method											
RS-mamba [7]	0.00	64.95	47.07	5.46	38.59	35.11	98.80	41.43	51.45	1172	0.27
Swin-UNet [21]	0.01	77.14	62.33	5.60	35.46	41.49	99.05	45.87	55.01	876	0.37
UNet [5]	0.00	77.05	65.46	12.14	51.91	44.08	99.01	49.95	59.55	1342	0.71
EMCAD [76]	6.95	79.29	65.94	29.70	45.90	50.75	98.80	53.90	65.19	1076	0.51
Segmenter [22]	12.70	74.49	60.48	40.66	40.20	52.35	98.89	54.26	66.66	**576**	0.55
DeepLabv3+ [73]	14.63	75.42	61.21	31.63	55.14	46.71	99.05	54.83	67.11	990	0.68
CCNet [77]	5.09	76.22	65.21	36.28	54.73	54.22	98.88	55.80	66.98	1096	0.56
CMTFNet [9]	12.23	79.14	67.61	37.12	53.77	56.66	98.83	57.91	69.52	980	0.45
PSPNet [75]	15.62	73.93	64.17	44.47	55.38	53.40	98.60	57.94	70.28	1070	0.72
DANet [74]	15.25	78.93	65.60	30.32	58.36	60.19	98.90	58.22	69.82	998	0.64
RS3Mamba [8]	11.16	80.49	69.97	35.31	54.47	57.34	99.06	58.26	69.53	1108	0.29
CTCFNet [29]	16.69	79.78	70.87	33.87	56.47	59.84	98.79	59.47	71.05	1394	0.20
SegFormer [6]	14.18	80.27	70.45	48.58	56.95	56.09	**99.08**	60.80	72.28	780	0.41
Memory-Efficient Methods											
CM-UNet [11]	18.41	81.06	74.93	45.48	63.85	59.83	98.87	63.20	74.44	4207	0.70
ICNet [33]	23.64	82.47	75.55	44.93	64.81	66.54	98.68	65.23	76.37	6919	1.71
UNetFormer [10]	27.32	**82.90**	76.33	42.95	64.79	66.12	99.00	65.63	76.85	4103	1.20
BiseNetv1 [12]	27.50	82.51	77.42	40.64	64.75	67.90	99.03	65.68	76.80	6929	1.41
GLNet [13]	22.95	79.84	76.25	51.69	62.22	67.96	99.04	65.71	76.85	2849	0.03
ISDNet [15]	24.14	79.81	74.52	54.00	65.65	68.39	98.75	66.47	77.58	2943	**1.87**
FCtL [14]	25.20	81.58	76.34	**57.73**	65.96	67.74	99.05	67.66	78.67	5682	0.01
MCPNet(Ours)	**28.26**	82.77	**77.84**	51.12	**69.35**	**68.83**	98.95	**68.16**	**78.96**	3318	1.19

omissions in detecting fragmented or isolated urban regions, our method significantly reduces misclassifications of other classes in areas where urban and other categories coexist. Overall, the visualizations in the first four rows highlight the model's superior capability in capturing fine structural details.

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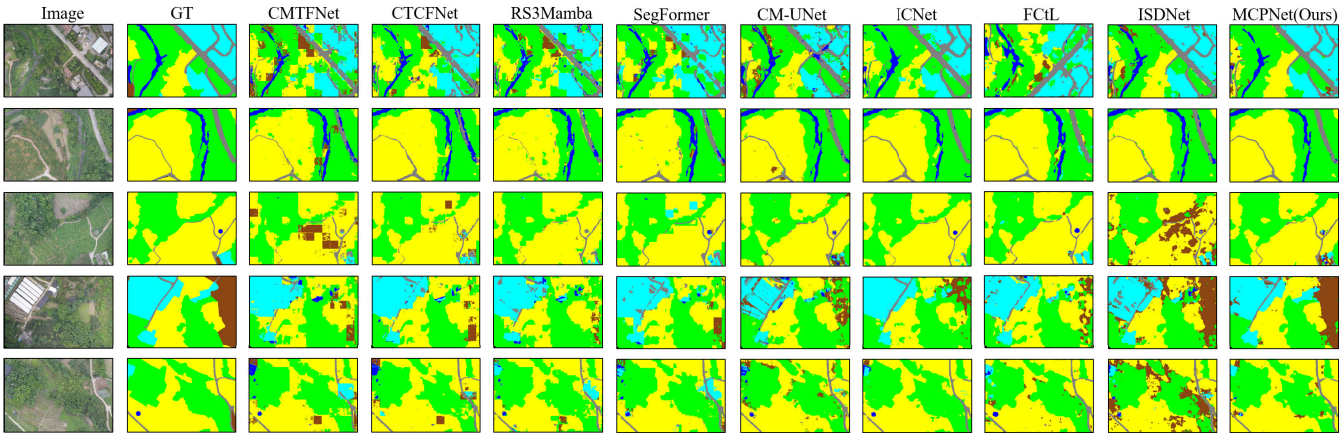


Fig. 8. Visualization results of several methods on the Yaan dataset, where yellow represents "farmland," gray represents "road," green represents "forest," brown represents "wasteland," cyan represents "built-up land," black represents "background," and blue represents "water."

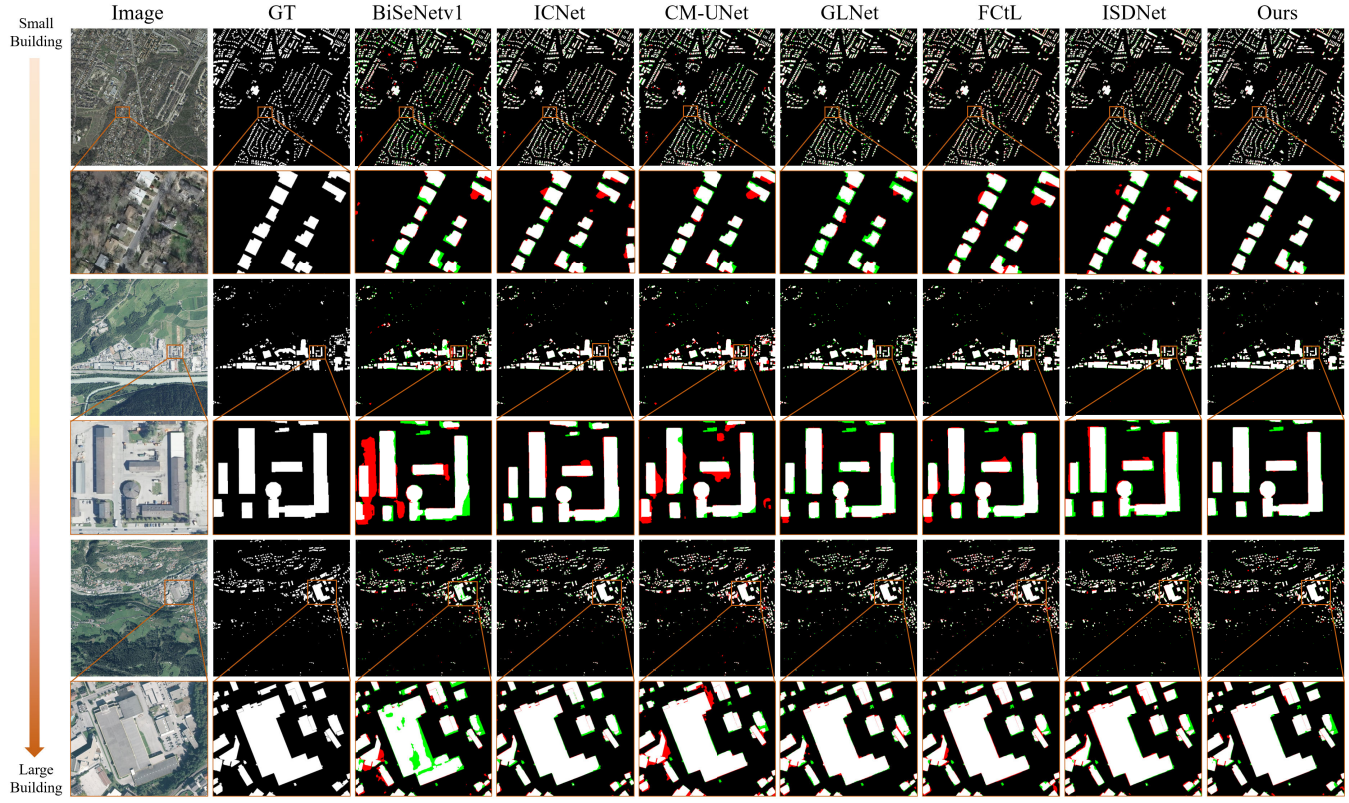


Fig. 9. Visualization examples of several methods on the Inria Aerial dataset. In the figure, white represents buildings, black represents the background, red indicates areas where the background is predicted as buildings, and green indicates areas where buildings are predicted as the background.

Next, the last four rows demonstrate the model's ability to classify large continuous landcover types. In the fifth, sixth, and seventh row scenes, significant gaps appear when using other methods. The internal pixels of "barren," "rangeland," and "forest" have high similarity, requiring the model to have a larger receptive field and utilize contextual semantics to distinguish different areas. Thanks to the global modeling capability of the M-branch, our method provides the most coherent semantic segmentation results. The final row shows a scene where large agricultural land areas include rangeland regions with urban areas. When the model lacks sufficient semantic information and the ability to extract detailed features, it is prone to confusing the low shrub boundary in the "rangeland" class with the distinct block structure of "agriculture," making the scene particularly challenging to distinguish. Our method leverages global semantics from the M-branch to guide local details and fuses them through the cascaded MG-FFM, effectively preserving both semantic and detailed information, resulting in the best semantic segmentation performance for this scene.

D. Evaluation on the Yaan Dataset

We apply MCPNet and the comparative methods to the Yaan UAV dataset, which features a higher spatial resolu-

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TABLE VI
QUANTITATIVE COMPARISON ON THE INRIA AERIAL DATASET. "†" DENOTES THE RESULTS COLLECTED FROM [20]

Method	IoU(%)	F1(%)	Memory(MB)	FPS
BiseNetv1 [12]	68.92	81.60	7179	1.48
CascadePSP †[16]	69.40	81.80	3236	/
GLNet [13]	70.31	82.56	**2850**	0.03
CM-UNet [11]	72.62	84.14	4285	0.75
FCtL [14]	73.08	84.44	5682	0.01
ISDNet [15]	73.55	84.76	3709	**1.90**
WSDNet †[20]	75.20	86.00	4379	/
ICNet [33]	75.34	85.94	7169	1.75
MCPNet(Ours)	**77.18**	**87.12**	3526	1.21

tion, richer image details, and more pronounced intra-class variations with smaller interclass differences, as illustrated in Figs. 1(a), 5(b), and 8. These characteristics impose greater demands on the global semantic modeling capability of the model. The experimental results are presented in Table V. The context loss caused by image patching limits the performance of conventional semantic segmentation methods, which can only achieve suboptimal semantic segmentation results. In contrast, memory-efficient methods that process larger image sizes directly capture more extensive contextual information, leading to superior performance. Among all methods, our approach attains the highest IoU scores on the "wasteland," "farmland," "road," and "built-up land" categories, with values of 28.26%, 77.84%, 69.35%, and 68.83%, respectively. For "forest" and "background," MCPNet also achieves performance close to the best results, at 82.77% and 98.95%. Although its performance on the "water" category slightly lags behind the best-performing method (FCtL), MCPNet still demonstrates a clear overall advantage when considering all of mIoU, mF1, and FPS. Overall, our method achieves improvements ranging from 0.5% to 26.73% in mIoU and from 0.29% to 27.51% in mF1 while also maintaining an efficient inference speed of 1.19 FPS when processing large-size images with an image size of 6000 × 4000.

Furthermore, we visualize the inference results on the Yaan dataset, as shown in Fig. 8. In the first and second rows, rivers and roads exhibit similarities in shape distribution and local texture color, while in the third to fifth rows, farmland and wasteland share similar characteristics. Without sufficient global semantic guidance, these land-cover classes tend to be misclassified. Additionally, our method produces the most continuous and smooth semantic segmentation results for roads and rivers across the five scenes.

E. Evaluation on the Inria Aerial Dataset

1) Comparative Experiment: To further demonstrate the effectiveness of our method, we apply it to the Inria Aerial dataset, where each image contains up to 25 million pixels, approximately four times the size of the DeepGlobe images. Additionally, the foreground regions in these images are more detailed, imposing higher demands on both semantic segmentation performance and memory usage. As shown in Table VI, our method achieves the best performance in terms of IoU and $F1$, outperforming the previous best methods by 1.84%–8.26% in IoU and 1.12%–5.52% in $F1$. Meanwhile, the inference speed of our method remains at a relatively high level.

Additionally, Fig. 9 displays the visualization results, where we show the results for dense small building clusters, medium-sized buildings, and large buildings in the first, third, and fifth rows. It is evident that our method produces more continuous and clearer boundaries within the extracted buildings. Compared to the other methods, our approach results in fewer instances of buildings being stuck together. Moreover, looking at the overall recognition results across the three scenes, our method also demonstrates a lower number of false positive and false negative pixels.

2) Model Cross-Domain Generalization Evaluation: To further evaluate the generalization capability of our method in complex geographic scenarios, we conducted cross-region generalization experiments on the Inria Aerial dataset. The dataset consists of five geographically distinct regions (Austin, Chicago, Kitsap County, Vienna, and West Tyrol), each exhibiting varying building densities (e.g., high density: Chicago and Vienna; low density: Kitsap County and West Tyrol) [31]. Each region contains 36 large-size images covering an area of 81 km^2, making the dataset particularly suitable for evaluating the cross-region generalization ability of semantic segmentation models. For simplicity, we refer to these regions as A, C, K, V, and T.

The experimental protocol is given as follows. In each experiment, four regions are selected to form the training set and the source domain test set, while the remaining region is used as the target domain for cross-region testing. This results in five experimental settings: "ACKT → V," "ACKV → T," "ACTV → K," "AKTV → C," and "CKTV → A." For each source region, the first 30 images are used for training, and the last six images are reserved for source domain testing. All 36 images from the target region are used for testing to thoroughly evaluate the model's generalization ability to unseen areas.

Table VII reports the performance of the building category in terms of IoU and $F1$-score across five cross-region tasks, for both the source domain (Src.) and target domain (Tgt.). The rightmost column, Overall, indicates the average performance across all cross-domain tasks (mIoU/mF1), serving as an overall assessment metric.

In terms of overall performance, our proposed MCPNet achieves the best results across all source domain tasks, with an mIoU of 76.75% and mF1 of 88.84%, significantly outperforming BiSeNetv1 (70.76%/82.86%) and CM-UNet (73.13%/84.38%). ISDNet also lags behind, with an average source domain IoU of 72.27%.

Regarding generalization to the target domain, MCPNet again demonstrates superior performance, achieving an average cross-domain IoU/F1 of 67.46%/80.43%. It consistently outperforms other methods across multiple tasks. For instance, in the "ACKV → T" task, MCPNet achieves a target domain IoU of 73.35%, compared to 67.32%, 70.98%, and 67.85% by BiSeNetv1, CM-UNet, and ISDNet, respectively. In the more challenging "AKTV → C" task, MCPNet attains a target IoU

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TABLE VII
CROSS-REGION GENERALIZATION COMPARISON OF DIFFERENT METHODS

Method	Domain	$ACKT \rightarrow V$		$ACKV \rightarrow T$		$ACTV \rightarrow K$		$AKTV \rightarrow C$		$CKTV \rightarrow A$		Overall	
		IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1	mIoU	mF1
BiSeNetv1 [12]	**Src.**	71.47	83.36	70.76	82.88	70.70	82.84	73.37	84.64	67.51	80.60	70.76	82.86
	Tgt.	67.18	80.37	67.32	80.47	56.45	72.16	56.82	72.47	66.17	79.64	62.79	77.02
CM-UNet [11]	**Src.**	74.01	85.06	74.96	85.69	76.23	86.51	76.91	86.95	63.53	77.70	73.13	84.38
	Tgt.	73.24	84.55	70.98	83.03	57.51	73.03	59.38	74.51	54.17	70.28	63.06	77.08
ISDNet [15]	**Src.**	74.59	85.45	71.75	83.55	70.92	82.99	73.14	84.49	70.94	83.00	72.27	83.90
	Tgt.	**74.54**	**85.41**	67.85	80.85	58.27	73.63	64.35	78.31	64.39	78.34	65.88	79.31
MCPNet(Ours)	**Src.**	**75.00**	**85.72**	**77.15**	**87.10**	**77.94**	**97.60**	**78.22**	**87.78**	**75.45**	**86.01**	**76.75**	**88.84**
	Tgt.	73.33	84.62	**73.35**	**84.63**	**58.54**	**73.85**	**65.40**	**79.08**	**66.66**	**79.99**	**67.46**	**80.43**

TABLE VIII
STRUCTURAL ABLATION EXPERIMENT ON THE DEEPGLOBE DATASET

Train-strategy	C-branch	M-branch	$\mathcal{G}_C$	$\mathcal{G}_R$	$\mathcal{G}_S$	mIoU(%) ↑	mF1(%) ↑	Tr.(bs=2) / Inf.(bs=1) Mem.(MB)↓	FPS↑
√	√					71.47	82.40	4979 / 2580	11.60
√		√				67.36	79.31	**2533 / 1280**	**13.83**
√	√	√	√			74.94	84.96	7329 / 2656	9.85
√	√	√	√	√		75.32	85.27	7329 / 2656	9.48
√	√	√			√	74.76	84.86	7335 / 2702	9.55
	√					67.59	79.20	16166 / 2580	11.60
		√				66.26	78.19	7471 / **1280**	**13.83**
	√	√	√			70.26	81.48	21061 / 2656	9.85
	√	√	√	√		70.96	81.94	21061 / 2656	9.48
	√	√			√	70.98	81.97	21077 / 2702	9.55
	√	√	√	√	√	71.21	82.05	22130 / 2794	9.20
√	√	√	√	√	√	**76.10**	**85.86**	7656 / 2794	9.20

of 65.40%, significantly outperforming BiSeNetv1 (56.82%) and CM-UNet (59.38%).

It is worth noting that in the "CKTV → A" task, the source domain IoU of BiSeNetv1 and CM-UNet drops to 67.51% and 63.53%, respectively, which is substantially lower than their performance in other tasks. This indicates a sensitivity to changes in the training region composition. In contrast, MCPNet maintains a source domain IoU of no less than 75% across all five experimental settings, highlighting its robustness and stronger adaptability to diverse regional combinations.

In summary, MCPNet not only maintains excellent semantic segmentation accuracy within individual regions but also demonstrates outstanding generalization and robustness in cross-region transfer tasks, confirming its broad applicability in remote sensing image semantic segmentation.

F. Ablation Study

1) Ablation Study of MCPNet Components: To validate the effectiveness of each component in MCPNet, we conducted detailed ablation experiments on the DeepGlobe dataset. The experimental results are presented in Table VIII, where a "√" symbol indicates the inclusion of the corresponding component. M-branch and C-branch denote training and testing performed exclusively with the Mamba branch and the CNN branch, respectively. The C-branch incorporates an auxiliary loss and employs a simple FCN head for decoding, while the M-branch retains the structural reconstruction process to preserve spatial structure and also utilizes an FCN head for decoding. Since MG-FFM is a multigated fusion module, the experiments progressively introduce each gating mechanism: $\mathcal{G}_C$ represents channel gating, $\mathcal{G}_R$ denotes residual gating, and $\mathcal{G}_S$ refers to spatial gating.

Please note that the experimental setup for the ablation study adheres to Section IV-B. In cases where memory overflow occurs during the experiments, we reduce the batch size to $1/n$ of the original configuration, adjust the initial learning rate to $1/n$ of its original value, and increase the number of iterations to n times the original setting.

The experimental results show that when only a single branch is included, the semantic segmentation performance is limited. The semantic segmentation performance of the C-branch is significantly better than that of the M-branch, with a 4.11% higher mIoU. This may be due to the downsampling in the M-branch, which results in a substantial loss of detailed information. The optimal results were achieved through the progressive fusion of the two branches using MG-FFM. Additionally, the semantic segmentation results were affected (by 0.78%–1.34%) when certain gating mechanisms were absent. The dynamic context training strategy, by incorporating multiscale and continuous contextual information, effectively enhances the model's adaptability to objects of varying scales in complex environments. This approach not only alleviates the memory constraints during training with large-size images but also significantly boosts model performance, with improvements ranging from 1.1% to 4.89% in mIoU and from 1.12% to 3.30% in mF1 while reducing training memory consumption by 4938–14 474 MB.

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TABLE IX
LOSS FUNCTION ABLATION STUDY ON THE DEEPGLOBE DATASET

$\mathcal{L}_{Seg}$	$\mathcal{L}_{AuxVss}$	$\mathcal{L}_{re}$	$\mathcal{L}_{aff}$	$\mathcal{L}_{Decoder}$	mIoU(%)	mF1(%)
√					74.68	84.88
√	√				74.95	85.05
√	√	√			75.15	85.12
√	√	√	√		75.36	85.34
√	√	√	√	√	**76.10**	**85.86**

TABLE X
EFFECT OF EACH VSS MODULE IN MCPNET

Method	mIoU(%)	mF1(%)	Memory(MB)	FPS
RSSG [39]	75.83	85.59	2904	9.05
VDB [80]	75.89	85.59	**2794**	9.24
LFSS [42]	75.45	85.33	2914	**9.32**
VSS [35]	**76.10**	**85.86**	**2794**	9.20

2) Ablation Study of Loss Function: Since our framework utilizes multiple auxiliary losses for joint training, we designed the loss ablation experiment shown in Table IX. The results in the table demonstrate that the model's performance gradually improves with the addition of each auxiliary loss.

Since MCPNet is a hybrid architecture of Mamba–CNN, we have designed a series of experiments to investigate the performance of VSS models on RGB images and validate the effectiveness of the detail preserving network and MG-FFM.

3) Ablation Study of VSS Module: For the first objective, we tested several VSS modules, including Residue State-Space Group (RSSG) [39], Vision Dehamba Block (VDB) [80], low-frequency state space (LFSS) [42], and VSS [35]. Based on the experimental results (as shown in Table X), we selected the VSS block as the foundational module for the M-branch, due to its superior accuracy and competitive inference speed (FPS).

4) Ablation Study of CNN Branch: For the second objective, we replaced the detail preservation network in the C-branch with several widely used convolutional neural networks for image processing tasks, including ResNet [81], EfficientNetV2 [82], MobileNetV2 [83], MobileNetV3 [84], ConvNeXt [60], PKINet [62], and SegNeXt [61]. Given that different networks have distinct structural designs, we followed a fundamental replacement principle to facilitate the experimental process. We retained the feature extraction layers of the network until a downsampling factor of 32. Similarly, when memory overflow occurs, we reduce the batch size to $1/n$ of the original configuration, adjust the initial learning rate to $1/n$ of its original value, and increase the number of iterations to n times the original setting. The experimental results, as shown in Table XI, indicate that our method achieved the highest scores in mIoU and $F1$, outperforming other networks by 0.8%–6.03% and 0.63%–4.34%. Additionally, the results suggest that a deeper or wider structure in the C-branch does not necessarily yield better performance. Under the same architecture, such as ResNet and PKINet, smaller networks not only exhibited lower memory usage but also demonstrated higher semantic segmentation accuracy. One possible reason for this is that an excessively heavy C-branch disrupts the balance between the C-branch and M-branch, thereby impairing model performance. Additionally, our method achieves the best semantic segmentation performance while maintaining a relatively high FPS.

TABLE XI
PERFORMANCE COMPARISON BETWEEN THE DETAIL PRESERVATION NETWORK AND OTHER NETWORKS

Method	mIoU(%)	mF1(%)	Memory(MB)	FPS
ResNet18 [81]	75.30	85.23	2120	**11.10**
ResNet50 [81]	74.71	84.82	2560	7.64
EfficientNetV2-M [82]	70.07	81.52	2768	6.52
MobileNetV2 [83]	74.96	85.09	2844	1.17
MobileNetV3 [84]	73.96	84.26	2890	9.59
ConvNeXt-T [60]	72.45	83.22	2992	7.36
PKINet-T [62]	73.02	83.73	**1854**	4.57
PKINet-S [62]	70.97	81.95	2772	9.46
SegNeXt-S [61]	73.07	83.69	2234	0.74
DPN(Ours)	**76.10**	**85.86**	2794	9.20

TABLE XII
COMPARISON OF DIFFERENT FUSION METHODS

Method	mIoU(%)	mF1(%)	Memory(MB)	FPS
FAM [29]	73.57	84.02	2848	8.77
GFM [85]	73.94	84.26	2850	10.15
CMXFFM [86]	70.91	82.08	**2668**	8.48
CFM [87]	73.07	83.58	2852	9.43
FFM [12]	74.19	84.48	2712	**10.93**
MG-FFM(Ours)	**76.10**	**85.86**	2794	9.20

5) Ablation Study of Fusion Module: To validate the effectiveness of our MG-FFM module, we conducted a comparative analysis of different feature fusion modules. Since these modules mismatch MCPNet, the features of both branches are uniformly processed as follows before applying each module. We unified the channel dimensions of the two-branch features using a 1×1 convolution and adjusted the spatial resolution of the low-resolution features using bilinear interpolation. Subsequently, we employed different fusion modules for feature integration. The experimental results, as shown in Table XII, indicate that our method achieves the best performance in terms of mIoU and $F1$-scores, outperforming the comparison methods by 1.91%–5.19% and 1.38%–3.78%. Compared to other fusion modules, our approach demonstrates significant improvements.

V. CONCLUSION

In this article, we first analyze the challenges faced by existing methods in semantic segmentation of large-size remote sensing images. To address these challenges, we propose an asymmetric Mamba–CNN collaborative architecture, MCPNet, specifically designed for this task. MCPNet constructs an M-branch based on SSMs, processing downsampled images to efficiently model global dependencies with linear complexity. The carefully designed DPN enhances the model's capability to capture fine details through a divergent downsampling strategy and multiscale convolutional kernel groups. The heterogeneous features from the two branches are progressively

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refined through an MG-FFM, achieving cross-modal feature alignment while filtering out redundant noise introduced by large-size images, thereby facilitating a progressive fusion of semantics and details.

Experiments conducted on the self-constructed Yaan UAV dataset and two public datasets demonstrate that our method achieves SOTA performance in terms of mIoU and $F1$-scores while requiring less than 3 GB of GPU memory to directly perform fine-grained semantic segmentation on a 6-million-pixel remote sensing image.

For future work, we will focus on the following directions.

1) Further exploration and optimization of SSMs for large-size remote sensing image semantic segmentation.
2) Integrating knowledge distillation techniques to further optimize the model structure and reduce memory consumption.

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